

Virocon Environmental Contour Analysis

A Python workflow for fitting a joint metocean model to **significant wave height** and **wave period** data, and then computing **environmental contours** for multiple return periods using **virocon**. The script supports both **omnidirectional** and **directional-sector** analyses, writes a **multi-page PDF** with contour plots, exports **high-resolution PNG figures**, and produces a detailed **results report** with fitted-model information and contour summaries.

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1. Purpose and scope

This repository contains a script that performs **environmental contour analysis** for oceanographic time series, with emphasis on the joint behaviour of:

- **Significant wave height Hs**
- **Peak wave period Tp**

- optionally **mean wave direction** MWD

Internally, the script converts T_p to **zero-upcrossing period** T_z and then fits a **global hierarchical model** using the predefined `get_OMAE2020_Hs_Tz()` model from `virocon`. After the model is fitted, it computes contours for one or more **return periods**, plots them, and reports the key design points such as:

- maximum H_s along each contour
- T_z associated with that maximum H_s
- T_p associated with that maximum H_s

This is a practical engineering script, but it is based on a substantial body of probabilistic extreme-value and structural-reliability theory.

2. Why environmental contours matter

In offshore and coastal engineering, one often needs a rational way to define **rare but plausible combinations** of environmental variables. A structure is usually not loaded by one variable in isolation. A vessel, jacket, monopile, floating unit, or breakwater experiences loads that depend on **combinations** of wave height, wave period, wind speed, current, and direction.

A traditional univariate extreme-value analysis can estimate, for example, a 100-year wave height. But it does **not** directly answer questions such as:

- What wave period tends to accompany very high waves?
- Which combinations of height and period are likely to be critical for response?
- How do those combinations depend on direction?
- Which multivariate design states should be selected for time-domain or structural simulations?

Environmental contours address this by converting a fitted **joint distribution** of environmental variables into a **curve of rare environmental states**. That curve can then be used as a screening surface for design load cases.

The environmental contour method is widely used because it is much cheaper than a full long-term response analysis, while still being rooted in structural reliability concepts.

3. Theoretical framework

3.1 Long-term extremes and return periods

A return period is a statistical statement, not a schedule. A **50-year contour** does **not** mean the contour is exceeded exactly once every 50 years. It means the contour is associated with an exceedance probability that is consistent, on average, with one exceedance in 50 years under the adopted discretization into environmental states.

If the environment is divided into sea states of duration t_s , and the target return period is T_R years, then the contour is linked to a small exceedance probability per sea state.

The script uses the standard relationship:

$$\alpha = \frac{t_s}{T_R \cdot 365.25 \cdot 24}$$

where:

- α = exceedance probability per sea state
- t_s = sea-state duration in hours
- T_R = return period in years

This is exactly the probability concept used by `virocon` through `calculate_alpha()`.

3.2 Environmental states and exceedance probability

The environmental contour method converts a **long-term** design target into a **per-state** exceedance probability. This is necessary because the metocean time series is treated as a sequence of environmental states, each assumed approximately stationary over a short duration.

For example, if:

- $t_s = 3 \text{ h}$
- $T_R = 50 \text{ years}$

then the total number of sea states in 50 years is approximately:

$$N = \frac{50 \cdot 365.25 \cdot 24}{3}$$

and the target exceedance probability per state is:

$$\alpha = \frac{1}{N}$$

The script lets `virocon` do this conversion consistently.

3.3 Joint probabilistic modelling of sea states

The central object in environmental contour analysis is the **joint probability distribution** of the environmental variables.

For two variables, here H_s and T_z , the joint density may be written as:

$$f_{H_s, T_z}(h, t) = f_{H_s}(h) \cdot f_{T_z|H_s}(t | h)$$

This factorization is important:

- $f_{H_s}(h)$ describes the long-term distribution of wave height
- $f_{T_z|H_s}(t | h)$ describes how wave period behaves **conditional on wave height**

This is more physically meaningful than assuming H_s and T_z are independent. In real sea states, wave period is typically dependent on wave height, and that dependence often matters strongly for dynamic response.

3.4 Global hierarchical models

`virocon` uses **global hierarchical models** as a central modelling approach.

A global hierarchical model is called:

- **global**, because it aims to describe the full range of the environmental variable rather than only a tail subset
- **hierarchical**, because variables are arranged in an ordered dependence structure

For a bivariate model:

1. the first variable is modelled marginally
2. the second variable is modelled conditionally on the first

For higher dimensions, each variable can depend only on variables earlier in the hierarchy.

This is one of the reasons the script first models wave height and then period conditional on wave height.

3.5 The Hs-Tz model used in this script

This script uses the predefined `virocon` joint model:

```
get_OMAE2020_Hs_Tz()
```

That predefined model comes from the OMAE 2020 work on global hierarchical models for wind and wave contours. In broad terms, it represents:

- Hs with an **Exponentiated Weibull** distribution
- Tz | Hs with a **conditional Lognormal** distribution

with the parameters of the conditional model controlled by **dependence functions** of Hs.

This model choice is important because:

- it is physically motivated
- it is already encoded in the upstream library
- it is documented in the `virocon` user guide
- it avoids the user having to manually rebuild a complex joint model

3.6 Why the script converts Tp to Tz

A very common practical issue is that measured or hindcast wave data often provide **peak period** T_p , whereas many joint wave-height/period contour models are formulated in terms of **zero-upcrossing period** T_z .

This script solves that mismatch by using a simple conversion:

$$T_z = \frac{T_p}{R_{Tp/Tz}}$$

where:

- $R_{\{Tp/Tz\}}$ is the configurable ratio `TP_TZ_RATIO`
- default value in the script is 1.2

This conversion is an engineering approximation. It is convenient and commonly used, but it is still an approximation. Therefore:

- the better your T_p/T_z conversion is for the local wave climate and spectral shape
- the better the resulting contour model is likely to represent the true sea-state statistics

3.7 Contour families and exceedance definitions

Environmental contour methods differ in how they define the **exceedance region**. This is not a minor implementation detail. It is one of the main reasons why different contour methods can produce materially different design states.

A useful high-level distinction is between methods based on:

- **marginal exceedance** concepts
- **total exceedance** concepts

Another distinction is whether the contour is constructed in:

- the **standard normal space**
- the **original physical variable space**

In the environmental contour benchmark literature, methods such as IFORM and direct sampling are usually grouped with **marginal exceedance** formulations, whereas ISORM and highest density are usually grouped with **total exceedance** formulations.

This matters because different exceedance definitions can yield noticeably different contours, even when they are built on the **same fitted joint distribution**.

3.8 IFORM, ISORM, highest density, and direct sampling

The script exposes four contour classes through the `CONTOUR_METHOD_TYPE` setting:

- IFORM
- ISORM
- HighestDensity
- DirectSampling

IFORM The **Inverse First-Order Reliability Method** is the default in the script. It is popular because it is standard, fast, and widely used in practice.

Conceptually, IFORM:

1. transforms the fitted joint distribution into a standard normal space
2. defines a reliability surface associated with the target exceedance probability
3. transforms that surface back to physical space

IFORM is a strong practical default and is often used for preliminary and design-stage contour work.

ISORM The **Inverse Second-Order Reliability Method** modifies the reliability approximation to better handle situations in which the first-order approximation may be non-conservative.

In the literature, ISORM is frequently described as a more conservative alternative, especially for some concave failure geometries in transformed space.

Highest Density A **highest density contour** encloses a region of probability mass such that the boundary corresponds to a constant joint density level.

In simple terms, it is the boundary of the most probable region containing the required non-exceedance probability.

This is conceptually different from IFORM and often yields different shapes, especially in asymmetric or strongly dependent distributions.

Direct Sampling The **direct sampling contour** avoids some of the approximations associated with transformed reliability spaces and works more directly in physical space via sampling-based procedures.

It can be attractive when one wants a contour construction method that stays closer to the original space and the model's own geometry.

3.9 Model fitting, parameter estimation, and dependence functions

A contour is only as credible as the fitted joint model beneath it.

In *virocon*, a joint model is built from:

- **distribution descriptions** (`dist_descriptions`)
- **fit descriptions** (`fit_descriptions`)
- **semantics** (`semantics`)

The upstream documentation explains these as follows:

- `dist_descriptions` define the joint model structure
- `fit_descriptions` define how parameter estimation is performed
- `semantics` define display metadata such as names, symbols, and units for plotting and labelling

In the current *virocon* documentation, the default behaviour is:

- **marginal distribution parameters** are estimated by **maximum likelihood estimation (MLE)**
- **dependence function parameters** are estimated by **least squares**
- weighted least squares can also be selected

This architecture is important because it cleanly separates:

1. **model structure definition**
2. **parameter estimation**

That separation is also one of the major architectural changes introduced in *virocon* 2.x.

3.10 Why directional sector analysis is useful

Many marine structures are strongly **direction-sensitive**.

Examples:

- a berth may be vulnerable to beam seas from only a narrow sector
- a rubble mound breakwater may have one exposed trunk and one sheltered trunk
- a vessel moored at a fixed heading may be much more sensitive to some approach sectors
- a coastal site may experience different swell systems from west, southwest, and south

For such problems, an omnidirectional contour is often too coarse. Directional analysis allows you to isolate the metocean statistics that matter for each approach sector.

This script therefore supports sector-based analysis using MWD, with user-controlled:

- sector width
- sector starting direction
- full circular coverage of 0–360°
- wrap-around sectors such as 340–10°

4. What this script does

At a high level, the script performs the following steps:

1. reads a CSV file with wave data
2. checks and cleans the data
3. optionally generates a dummy `input.csv` if no file is found
4. converts `Tp` to `Tz`
5. loads the predefined OMAE 2020 `Hs-Tz` model from `virocon`
6. fits the model to the data
7. retries fitting using more robust settings if needed
8. falls back to a simpler independent model if the complex fit fails
9. computes contours for all requested return periods
10. plots and exports the results
11. optionally repeats the analysis for directional sectors
12. writes a text report summarizing settings, fitted models, and contour maxima

5. Workflow implemented by the script

The script follows the same broad workflow promoted by the `virocon` user guide:

1. **Load environmental data**
2. **Define or import a joint model**
3. **Fit the model**
4. **Choose return period and sea-state duration**
5. **Compute contour**
6. **Plot contour and inspect results**

The script automates these steps in a reproducible engineering workflow and extends them with:

- robust fit retries
- sector analysis
- configuration logging
- PNG/PDF/report generation

6. Input data requirements

The script expects a CSV file, by default:

`input.csv`

Required columns

Column	Meaning	Units	Required
<code>datetime</code>	timestamp	date-time	yes
<code>swh</code>	significant wave height Hs	m	yes
<code>pp1d</code>	peak period Tp	s	yes
<code>mwd</code>	mean wave direction	degrees	only if sector analysis is enabled

Expected format

- `datetime` should be parseable by **pandas**
- **Hs** must be positive for valid modelling
- **Tp** must be positive for valid modelling
- **MWD** should normally be expressed in degrees in the 0–360 convention

Important note on direction convention

The script assumes that the **MWD** column is already consistent with the engineering interpretation you want to analyse. In practical datasets, mean wave direction may be reported as:

- **coming from** a direction
- **going to** a direction

The script does not transform conventions automatically. You must ensure the source convention matches your design interpretation.

7. Installation

Python environment

The upstream **virocon** repository currently recommends Python:

- 3.11
- 3.12
- 3.13

Older versions may work, but upstream states they are not actively tested.

Minimal installation for this script

```
pip install pandas numpy matplotlib virocon
```


Recommended: create a clean virtual environment

Windows

```
python -m venv .venv
.venv\Scripts\activate
pip install --upgrade pip
pip install pandas numpy matplotlib virocon
```

Linux / macOS

```
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install pandas numpy matplotlib virocon
```

Install virocon from PyPI

```
pip install virocon
```

Install virocon from the upstream GitHub main branch

```
pip install https://github.com/virocon-organization/virocon/archive/main.zip
```

Install virocon via conda-forge

```
conda create --name virocon python=3.13
conda activate virocon
conda install -c conda-forge virocon
```

8. Detailed configuration guide

The script exposes a user configuration block near the top of the file.

File and column settings

```
INPUT_FILE = "input.csv"
TIME_COL = "datetime"
HS_CSV_COL_NAME = "swH"
TP_CSV_COL_NAME = "pp1d"
MWD_CSV_COL_NAME = "mwd"
```

These parameters define:

- which CSV file is read
- which column is interpreted as time
- which columns contain Hs, Tp, and MWD

Tp-to-Tz conversion

```
TP_TZ_RATIO = 1.2
```

Used as:

```
Tz = Tp / TP_TZ_RATIO
```

Sector analysis settings

```
PERFORM_SECTOR_ANALYSIS = True
SECTOR_WIDTH_DEGREES = 30
START_DIR_DEGREES = 0
```

This part controls whether directional analysis is performed, the width of each sector, and the starting azimuth of the sectorization.

Examples:

- SECTOR_WIDTH_DEGREES = 30, START_DIR_DEGREES = 0
sectors: 0-30, 30-60, 60-90, ..., 330-0
- SECTOR_WIDTH_DEGREES = 30, START_DIR_DEGREES = 10
sectors: 10-40, 40-70, 70-100, ..., 340-10

The sector builder always covers the full 360° circle. If the width does not divide 360 exactly, the final sector is shortened so that there is no angular gap.

Contouring parameters

```
SEA_STATE_DURATION_HOURS = 3.0
RETURN_PERIODS_YEARS = [1, 5, 10, 25, 50, 100, 250]
MIN_SAMPLES_FOR_FIT = 200
```

- SEA_STATE_DURATION_HOURS controls the environmental-state duration used in the exceedance probability
- RETURN_PERIODS_YEARS is the list of contours to compute
- MIN_SAMPLES_FOR_FIT prevents fitting a model to a statistically too-small sample or sector

Plotting and outputs

```
SWAP_AXES_CONTOUR_PLOT = True
CONTOUR_METHOD_TYPE = "IFORM"
PDF_OUTPUT_FILE = "contours.pdf"
PNG_OUTPUT_DIR = "plots"
RESULTS_TXT_FILE = "results.txt"
```

- SWAP_AXES_CONTOUR_PLOT = True means period goes on the x-axis and Hs on the y-axis
- CONTOUR_METHOD_TYPE selects the contour family
- the remaining variables control output filenames and directories

Robust fitting strategies

The script defines a list of progressive fitting strategies:

1. default fitting
2. perturbed initial guesses (p0)
3. widened parameter bounds

This is especially useful when sectors have:

- small sample counts
- unusual data ranges
- highly clustered or narrow distributions

- optimization difficulties
-

9. Step-by-step usage

Step 1 — Prepare the input CSV

Create `input.csv` in the same directory as the script, unless you change `INPUT_FILE`.

Example:

```
datetime,swl,pp1d,mwd
2023-01-01 00:00:00,2.10,8.50,245
2023-01-01 01:00:00,2.25,8.70,248
2023-01-01 02:00:00,1.95,8.10,238
2023-01-01 03:00:00,3.40,9.60,255
```

If `PERFORM_SECTOR_ANALYSIS = False`, the script only requires:

- `datetime`
- `swl`
- `pp1d`

Step 2 — Adjust configuration

Open the script and check:

- file name and column names
- `TP_TZ_RATIO`
- sector settings
- sea-state duration
- return periods
- contour method
- output filenames

Step 3 — Run the script

```
python script.py
```

Step 4 — Inspect console messages

The script prints progress messages such as:

- loading data
- fitting attempt number
- fitting strategy used
- sector currently being analysed
- PNG save status
- report-writing status

Step 5 — Review generated outputs

After a successful run, review:

- `contours.pdf`

- `plots/*.png`
- `results.txt`

Step 6 — Check whether a fallback model was used

Read `results.txt`. The report states whether the primary OMAE 2020 model fit succeeded or whether the script had to fall back to a simplified independent model.

This is important for engineering interpretation.

10. Directional sector logic

The script contains explicit helper functions to build and apply sectors.

Angle normalization

All directions are normalized to the interval:

`[0, 360)`

Sector construction

The sector builder creates sectors starting at:

`START_DIR_DEGREES % 360`

then advances by `SECTOR_WIDTH_DEGREES` until the full circle is covered.

Interval convention

The script uses the convention:

- **lower bound inclusive**
- **upper bound exclusive**

So a sector 30-60 means:

`30° <= direction < 60°`

Wrap-around sectors

Wrap-around sectors are handled correctly.

Example:

`340-10`

means:

`direction >= 340° OR direction < 10°`

Full-circle special case

If:

`SECTOR_WIDTH_DEGREES = 360`

the script generates one full-circle sector.

Why this matters

This sector logic avoids common directional-analysis bugs such as:

- missing the 350–0–10 interval
 - inconsistent wrap-around handling
 - leaving gaps when the width does not divide 360 exactly
-

11. Outputs generated by the script

11.1 `contours.pdf`

A multi-page PDF containing:

- omnidirectional contour plots
- one page per directional sector if sector analysis is enabled
- all selected return periods overlaid
- a small table summarizing the maximum Hs and corresponding periods

11.2 `plots/`

The script exports a high-resolution PNG for each analysis case:

- omnidirectional
- each directional sector

These are useful for:

- reports
- slide decks
- design notes
- comparison between sectors

11.3 `results.txt`

This is the most important textual output. It includes:

- an overall summary table
 - configuration values used in the run
 - number of loaded points
 - fitted model representation
 - fit-attempt history
 - contour summary table for each analysis case
 - warning and error notes if any sectors were skipped
-

12. Robustness strategy and fallback modelling

Real metocean data are messy. Directional subsets can be sparse, tails can be difficult, and numerical optimization can fail.

The script therefore includes three layers of robustness.

12.1 Positive-value filtering

After converting T_p to T_z , the script filters out data where:

- $H_s \leq 0$
- $T_z \leq 0$

This is necessary because such values are physically invalid for the chosen distributions and can break log-based models.

12.2 Multiple fitting attempts

The script tries the full OMAE 2020 model using several strategies.

Strategy A — default parameters The first fit attempt uses the standard upstream fitting description.

Strategy B — perturbed initial guesses If the first attempt fails, the script perturbs initial values for dependence-function parameters. This can help the optimizer escape a poor starting region.

Strategy C — wider bounds If necessary, the script expands parameter bounds. This widens the permissible search space and can help convergence in difficult sectors.

12.3 Simplified fallback model

If all full-model attempts fail, the script builds a simpler model in which:

- H_s remains marginal
- T_z is treated as marginal as well
- the conditional dependence is removed

This is statistically less rich, but it allows the workflow to complete and gives the user a usable contour instead of a total failure.

Engineering meaning of the fallback

A fallback contour is **not equivalent** to a fully fitted dependent H_s - T_z model. It should be interpreted as a robustness measure, not as the preferred scientific result.

When a fallback occurs, you should consider:

- whether the sector has too few data
 - whether the period conversion is reasonable
 - whether the sector width should be changed
 - whether the dataset should be extended
 - whether a custom model would fit better than the predefined model
-

13. Interpretation of results

Maximum Hs along the contour

The script reports the largest wave height found on each contour. This is often a useful first screening metric, but it is **not automatically the governing structural state**.

A structure may instead be governed by:

- a lower Hs paired with a longer period
- a directional combination
- a resonance-sensitive period range
- a load component more sensitive to steepness than height alone

Tz and Tp at the point of maximum Hs

The script reports both:

- Tz @ max Hs
- Tp @ max Hs

This is useful because:

- the fitted model is built in Hs-Tz
- many engineering analyses are still easier to communicate and run in Hs-Tp

Omnidirectional versus sectoral contours

Use the omnidirectional contour when you need a broad all-directions summary.

Use sectoral contours when:

- structure orientation matters
- shielding or fetch effects matter
- local exposure differs by heading
- design checks are direction-specific

Comparing contour methods

A contour from IFORM should not be expected to exactly match a contour from HighestDensity or ISORM, even when fitted to the same data.

Differences can arise from:

- different exceedance-region definitions
- transformed-space versus physical-space construction
- local sensitivity to tail fitting
- dependence modelling choices

14. Engineering assumptions, limitations, and cautions

This script is useful, but it does not remove engineering judgement.

14.1 It is not a full long-term response analysis

Environmental contours are approximate design tools. They do **not** fully account for the complete response distribution conditional on every environmental state.

14.2 Tp-to-Tz conversion is approximate

The script assumes a constant Tp/Tz ratio. In reality, the relationship depends on spectral shape and sea-state characteristics.

14.3 Joint model choice matters

Published benchmark work shows that differences in the **joint distribution model** can drive larger contour differences than the choice of contour-construction algorithm itself.

14.4 Directional data convention matters

Do not mix “coming from” and “going to” conventions without explicitly correcting the data.

14.5 Minimum sample count is a practical threshold, not a theorem

MIN_SAMPLES_FOR_FIT = 200 is a pragmatic safeguard. It does not guarantee a good fit; it only helps avoid obviously under-informed fits.

14.6 The contour is not the response envelope

The response-governing state may lie at one point on the contour, but which point that is depends on the structural response model.

14.7 Sector width is a modelling choice

Narrow sectors improve directional resolution but reduce sample size. Wider sectors improve statistical stability but smear directional specificity.

15. Extending or adapting the script

There are several natural extensions.

Add wind or current

virocon supports higher-dimensional modelling. A future version of the script could extend from Hs-Tz to:

- V-Hs
- V-Hs-Tz
- Hs-Tz-U
- V-Hs-Tz-current

provided the user defines or imports an appropriate joint model.

Replace the Tp/Tz conversion

If your dataset or standards framework provides a more defensible transformation between Tp and Tz, replace the constant ratio.

Use custom joint models

The `virocon` user guide includes detailed examples of how to define:

- custom dependence functions
- custom distributions
- custom fitting descriptions
- semantics dictionaries for plotting

Add goodness-of-fit diagnostics

A future enhancement could export:

- marginal quantile plots
- dependence-function plots
- sector-wise fit diagnostics
- fit-quality metrics

Add extreme-point extraction rules

Right now the script reports the point of maximum Hs. Some applications may also want:

- maximum Tp
- maximum steepness
- percentile-based contour stations
- equally spaced arc-length stations

16. Troubleshooting

ModuleNotFoundError: No module named 'virocon'

Install dependencies:

```
pip install pandas numpy matplotlib virocon
```

CSV columns not found

Check the configuration block:

```
TIME_COL
HS_CSV_COL_NAME
TP_CSV_COL_NAME
MWD_CSV_COL_NAME
```

Make sure the CSV header exactly matches these names.

Sector analysis fails

Check:

- MWD column exists
- directions are numeric
- directions use a consistent convention
- `SECTOR_WIDTH_DEGREES > 0`
- `SECTOR_WIDTH_DEGREES <= 360`

Many sectors are skipped

This usually means too few data fell into those sectors. Options:

- widen sectors
- reduce `MIN_SAMPLES_FOR_FIT`
- increase record length
- disable sector analysis

Fitting repeatedly fails

Possible causes:

- poor data quality
- strong outliers
- very sparse sectors
- inappropriate `Tp/Tz` conversion
- model mismatch

Possible responses:

- inspect the data
- try wider sectors
- test a different contour method
- define a custom model in `virocon`
- accept the fallback model only as a temporary diagnostic result

The dummy `input.csv` was created and the script exited

This is expected. Populate the generated file with real data, then run the script again.

17. Upstream `virocon` context

The script sits on top of `virocon`, so it is useful to understand what the upstream library provides.

What `virocon` is

`virocon` is an importable Python package for computing environmental contours. The upstream repository describes it as a package intended to support the design of marine structures subjected to wave, wind, and current load combinations.

Methods implemented upstream

The current upstream repository states that `virocon` implements:

- joint probability modelling with a global hierarchical model
- parameter estimation for that model
- contour calculation using:
 - IFORM
 - ISORM
 - direct sampling
 - highest density
- “AND” and “OR” exceedance contours

Architectural context

The 2022 software update paper explains the major version-2 architecture:

- defining model structure and fitting are separated
- users can define dependence functions freely
- a `predefined.py` module exposes ready-made joint models

This architecture aligns very directly with how this script works:

1. import predefined model
2. instantiate `GlobalHierarchicalModel`
3. fit it
4. compute contours

User guide structure

The upstream user guide contains sections on:

- installation
- quick start
- overall workflow
- useful contour definitions
- detailed examples
- predefined and custom joint models
- parameter estimation
- contour construction
- the `dist_descriptions`, `fit_descriptions`, and `semantics` structures

That guide is the best starting point for users who want to move from this script to more custom `virocon` workflows.

18. Detailed bibliography

This section intentionally combines:

1. **software and documentation resources**
2. **technical and scientific references** relevant to environmental contour analysis

18.1 Software, repository, and documentation resources

1. **virocon-organization.** *virocon: A Python package to compute environmental contours.*
GitHub repository.
URL: <https://github.com/virocon-organization/virocon>
Relevance: official upstream source code, installation notes, examples, implemented methods, licensing, and release history.
2. **virocon documentation.** *User guide.*
URL: https://virocon.readthedocs.io/en/latest/user_guide.html
Relevance: official usage guide for the newest documented version, including installation, workflow, model structures, parameter estimation, and contour examples.
3. **virocon documentation.** *Quick start example: Sea state contour.*
URL: <https://virocon.readthedocs.io/en/latest/example.html>
Relevance: concise worked example showing the standard workflow of loading data, importing `get_OMAE2020_Hs_Tz()`, fitting `GlobalHierarchicalModel`, computing `alpha`, building an `IFORMContour`, and plotting.
4. **virocon documentation.** *Overall workflow and software architecture.*
URL: <https://virocon.readthedocs.io/en/latest/workflow.html>
Relevance: conceptual flowchart and class-level explanation of how model fitting and contour construction relate.
5. **virocon documentation.** *Statistical distributions and parameter estimation.*
URL: https://virocon.readthedocs.io/en/latest/joint_models_estimating_parameters.html
Relevance: explains predefined joint models, custom models, and fitting defaults such as MLE for marginals and least squares or weighted least squares for dependence functions.
6. **virocon documentation.** *Cheat sheet for the “description” data structures.*
URL: https://virocon.readthedocs.io/en/latest/description_cheat_sheet.html
Relevance: practical reference for `dist_descriptions`, `fit_descriptions`, and `semantics`.

18.2 Core software papers on virocon

7. **Haselsteiner, A.F., Lehmkuhl, J., Pape, T., Windmeier, K.-L., Thoben, K.-D. (2019).**
ViroCon: A software to compute multivariate extremes using the environmental contour method.
SoftwareX, 9, 95–101.
DOI: <https://doi.org/10.1016/j.softx.2019.01.003>
Relevance: original ViroCon software paper; explains motivation, implemented contour methods, and the original software architecture.
8. **Haselsteiner, A.F., Windmeier, K.-L., Ströer, L., Thoben, K.-D. (2022).**
Update 2.0 to “ViroCon: A software to compute multivariate extremes using the environmental contour method”.
SoftwareX, 20, 101243.
DOI: <https://doi.org/10.1016/j.softx.2022.101243>
Relevance: explains the major redesign in virocon 2.x, especially separation of model definition and fitting, user-definable dependence functions, and predefined model structures.

9. **Windmeier, K.-L., Haselsteiner, A.F., Ströer, L., Thoben, K.-D. (2021).**

Using virocon to calculate environmental contours for offshore wind turbine design.
WESC 2021 presentation.

Relevance: a compact practical presentation of the upstream workflow, especially for engineering users who want a task-oriented introduction.

18.3 Environmental contour methodology, reviews, and benchmark studies

10. **Haselsteiner, A.F., Coe, R.G., Manuel, L., Chai, W., Leira, B., Clarindo, G., Guedes Soares, C., Hannesdóttir, Á., Dimitrov, N., Sander, A., Ohlendorf, J.-H., Thoben, K.-D., de Hauteclocque, G., Mackay, E., Jonathan, P., Qiao, C., Myers, A., Rode, A., Hildebrandt, A., Schmidt, B., Vanem, E., Huseby, A.B. (2021).**

A benchmarking exercise for environmental contours.

Relevance: major comparison study showing that contour results depend both on the fitted joint distribution and on the contour-construction method; highly relevant for understanding uncertainty and method sensitivity.

11. **Ross, E., Astrup, O.C., Bitner-Gregersen, E., Bunn, D., Feld, G., Gouldby, B., Huseby, A.B., Liu, Y., Randell, D., Vanem, E., Jonathan, P. (2019).**

A review of environmental contour methods for marine and coastal design.

Relevance: broad review paper on the field; useful for understanding the method family, conceptual foundations, and practical limitations.

12. **Eckert, J., Martin, N., Coe, R., Seng, B.E., Stuart, J., Morrell, A. (2021).**

Comparison framework for environmental contours.

Relevance: supports systematic method comparison and evaluation.

18.4 Foundational contour and reliability references

13. **Haver, S. (1985).**

Early work on environmental contour concepts and constant exceedance formulations.

Relevance: foundational source in the historical development of the method.

14. **Haver, S. (1987).**

Further development of environmental contour concepts.

Relevance: historical continuation of early contour methodology.

15. **Winterstein, S.R., Ude, T.C., Cornell, C.A., Bjerager, P., Haver, S. (1993).**

Environmental parameters for extreme response: inverse FORM with omission factors.

Relevance: seminal IFORM reference and one of the most important papers in the practical environmental contour tradition.

16. **Chai, W., Leira, B.J. (2018).**

ISORM-based environmental contour formulation.

Relevance: important reference for second-order reliability-based contour construction.

17. **Huseby, A.B., Vanem, E., Natvig, B. (2013).**

Direct sampling contour method.

Relevance: key reference for sampling-based contour construction in physical space.

18. **Haselsteiner, A.F., Ohlendorf, J.-H., Wosniok, W., Thoben, K.-D. (2017).**

Highest density contour method.

Relevance: important reference for density-level-set contours with total exceedance interpretation.

19. **Dimitrov, N. (2020).**

Inverse directional simulation contours.

Relevance: reference for directional simulation contour methodology and its relation to ISORM.

20. **Derbanne, Q., de Hauteclocque, G. (2019).**

Direct IFORM.

Relevance: connects reliability ideas with more direct construction in physical space.

18.5 Joint distribution modelling and dependence structure references

21. **Mathisen, M., Bitner-Gregersen, E. (1990).**

Early hierarchical or conditional metocean modelling ideas.

Relevance: historical background for conditional wave-variable modelling.

22. **Bitner-Gregersen, E. (2015).**

Advanced joint wave and wind distribution modelling.

Relevance: important practical and theoretical reference for offshore environmental modelling.

23. **Horn, J.T., Bitner-Gregersen, E., Krokstad, J.R., Leira, B.J., Amdahl, J. (2018).**

Global hierarchical-type modelling approaches for metocean variables.

Relevance: useful background for model-structure design.

24. **Cheng, W., Svangstu, E., Moan, T., Gao, Z. (2019).**

Further developments in joint metocean modelling for contour work.

Relevance: modern comparison point for hierarchical modelling strategies.

25. **Haselsteiner, A.F., Sander, A., Ohlendorf, J.-H., Thoben, K.-D. (2020).**

Global hierarchical models for wind and wave contours: physical interpretations of the dependence functions.

Proceedings of OMAE 2020.

DOI: <http://dx.doi.org/10.1115/OMAE2020-18668>

Relevance: central model reference for the predefined OMAE2020 models used by *virocon*, including the Hs-Tz model used by this script.

26. **Haselsteiner, A.F., Thoben, K.-D. (2020).**

On fitting the exponentiated Weibull distribution to significant wave height data with emphasis on the upper tail.

Relevance: explains why exponentiated Weibull can be attractive for Hs modelling and how tail-focused estimation may improve engineering usefulness.

18.6 Alternative multivariate extreme-model families

27. **Vanem, E. (2016).**

Copula-based multivariate metocean modelling.

Relevance: alternative to hierarchical models.

28. **Fazeres-Ferradosa, T., Taveira-Pinto, F., Vanem, E., Reis, M.T., das Neves, L. (2018).**
Copula approaches in marine and coastal extreme analysis.
Relevance: useful for comparing modelling philosophies.
29. **Manuel, L., Nguyen, P.T.T., Canning, J., Coe, R., Eckert-Gallup, A., Martin, N. (2018).**
Joint extreme modelling alternatives for offshore applications.
Relevance: comparison with non-hierarchical models.
30. **Jonathan, P., Flynn, J., Ewans, K. (2010).**
Conditional extremes models for metocean variables.
Relevance: important alternative tail-modelling strategy.
31. **Jonathan, P., Ewans, K., Flynn, J. (2014).**
Joint exceedance and conditional extreme contour-related approaches.
Relevance: relevant for tail-focused design state estimation.

18.7 Guidance and standards

32. **DNV GL (2017).**
DNVGL-RP-C205: Environmental conditions and environmental loads.
Relevance: one of the most influential offshore engineering guidance documents for environmental loading and joint environmental models.
33. **NORSOK (2007).**
N-003: Actions and action effects.
Relevance: guidance on design loading context, including practical safety handling.
34. **International Electrotechnical Commission (2019).**
IEC 61400-3-1: Design requirements for fixed offshore wind turbines.
Relevance: important standard in offshore wind design where environmental contours are commonly used.

18.8 Response-based and full long-term analysis context

35. **Guedes Soares, C. (1993).**
Full long-term response analysis concepts.
Relevance: foundational reference for the more complete alternative to contour-based screening.
36. **Muliawan, M.J., Gao, Z., Moan, T. (2013).**
Response estimation through full long-term approaches.
Relevance: useful contrast against environmental contour approximations.
37. **Vanem, E., Guo, B., Ross, E., Jonathan, P. (2020).**
Response-based long-term analysis in offshore reliability.
Relevance: important reminder that contour methods are approximations, not substitutes in all contexts.